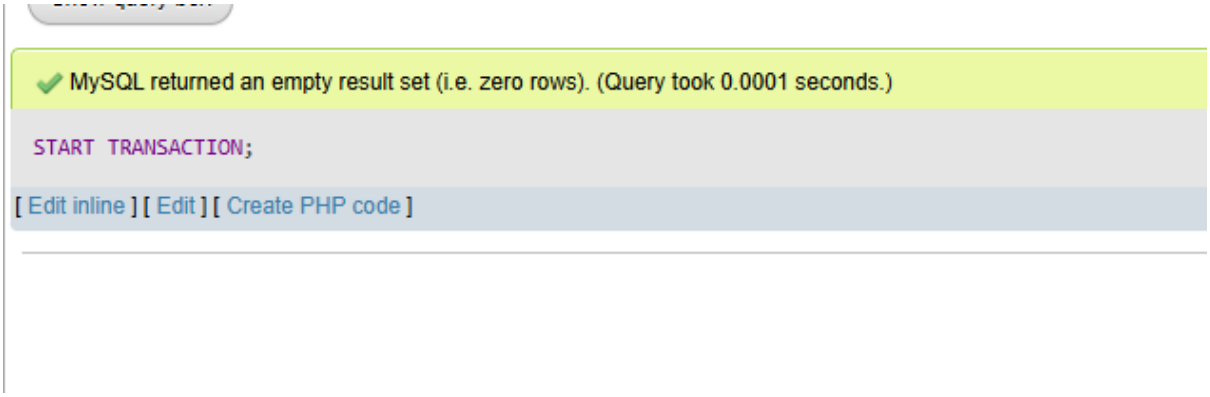
1.

START TRANSACTION;

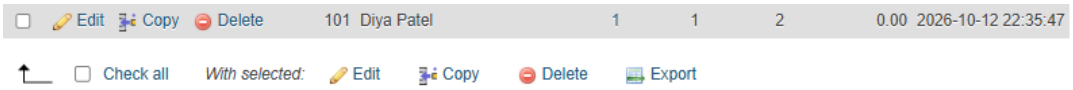


INSERT INTO Orders

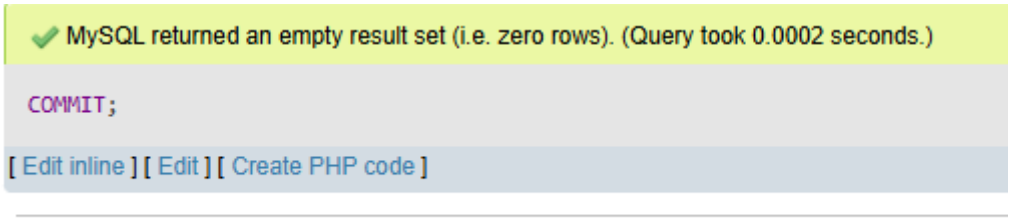
(order_id, customer_name, restaurant_id, item_id, quantity)

VALUES

(101, 'Diya Patel', 1, 1, 2);



COMMIT;

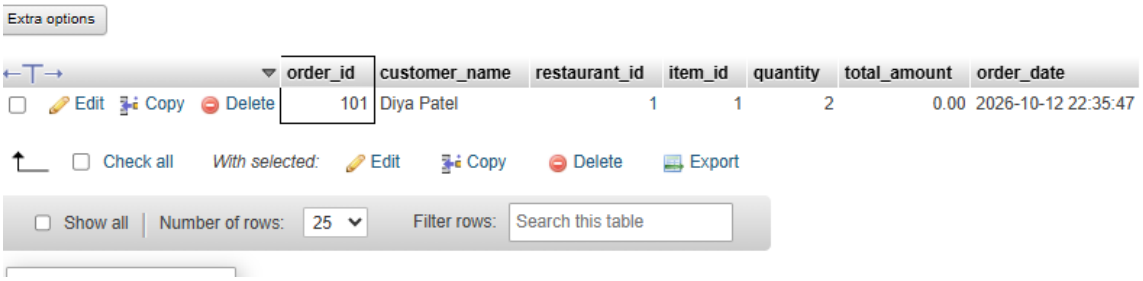


Verify the Order

SELECT *

FROM Orders

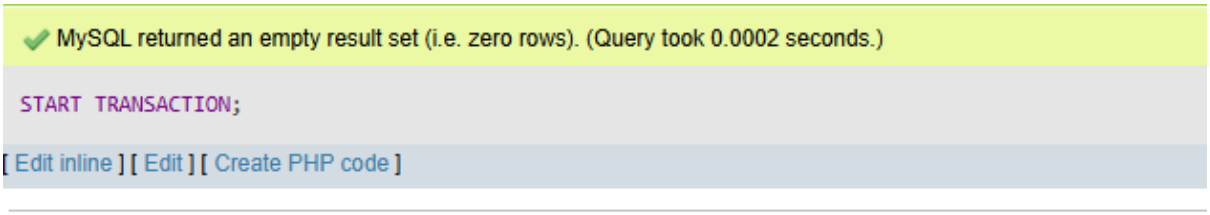
WHERE order_id = 101;



2.

Simulate a Zomato-style food order process: insert two new items into an OrderItems table, then use ROLLBACK to undo the changes before committing. Check that no new items remain in the table after rollback.

START TRANSACTION;



INSERT INTO OrderItems

(order_item_id, order_id, item_name, quantity)

VALUES

(201, 101, 'Pizza', 1),

(202, 101, 'Burger', 2);

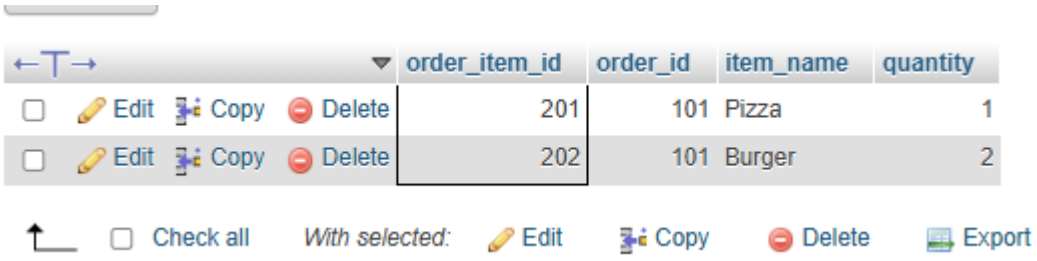


ROLLBACK;

SELECT *

FROM OrderItems

WHERE order_item_id IN (201, 202);



	order_item_id	order_id	item_name	quantity
Edit Copy Delete	201	101	Pizza	1
Edit Copy Delete	202	101	Burger	2

Check all With selected: Edit Copy Delete Export

3.

In a Flipkart-like shopping cart scenario, use SAVEPOINT to mark a point after adding a product to the Cart table, add another product, then use ROLLBACK TO SAVEPOINT to undo only the last addition. Show the final contents of the Cart table.

```
CREATE TABLE Cart (
    cart_id INT PRIMARY KEY,
    product_name VARCHAR(100),
    quantity INT,
    price DECIMAL(10,2)
) ENGINE=InnoDB;


START TRANSACTION;


INSERT INTO Cart
(cart_id, product_name, quantity, price)
VALUES
(1, 'Headphones', 1, 999.00);
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0002 seconds.)

```
START TRANSACTION;
```

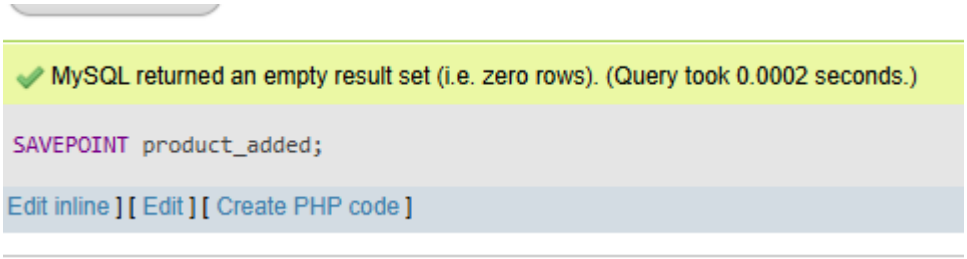
Edit inline] [Edit] [Create PHP code]

1 row inserted. (Query took 0.0006 seconds.)

```
INSERT INTO Cart (cart_id, product_name, quantity, price) VALUES (1, 'Headphones', 1, 999.00);
```

Edit inline] [Edit] [Create PHP code]

```
SAVEPOINT product_added;
```



INSERT INTO Cart

(cart_id, product_name, quantity, price)

VALUES

(2, 'Smart Watch', 1, 1999.00);

ROLLBACK TO SAVEPOINT product_added;

4. Write a trigger on a Wallet table that automatically deducts the purchase amount from the user's balance whenever a new transaction is inserted, similar to how Paytm updates wallet balance after a payment.

```
CREATE TABLE Wallet (
    user_id INT PRIMARY KEY,
    user_name VARCHAR(100),
    balance DECIMAL(10,2)
) ENGINE=InnoDB;

INSERT INTO Wallet (user_id, user_name, balance)
VALUES (1, 'Diya Patel', 5000.00);

CREATE TABLE Transactions (
    transaction_id INT PRIMARY KEY,
    user_id INT,
    amount DECIMAL(10,2),
    FOREIGN KEY (user_id) REFERENCES Wallet(user_id)
) ENGINE=InnoDB;
```

CREATE TRIGGER deduct_wallet_balance

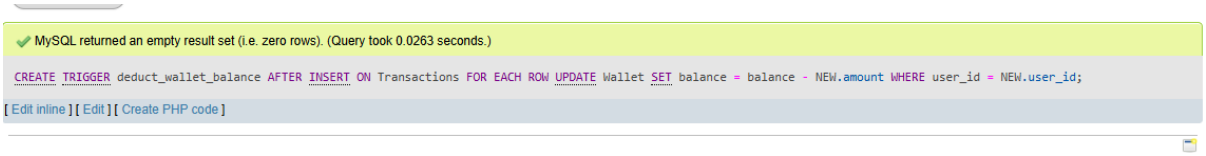
AFTER INSERT ON Transactions

FOR EACH ROW

UPDATE Wallet

SET balance = balance - NEW.amount

WHERE user_id = NEW.user_id;



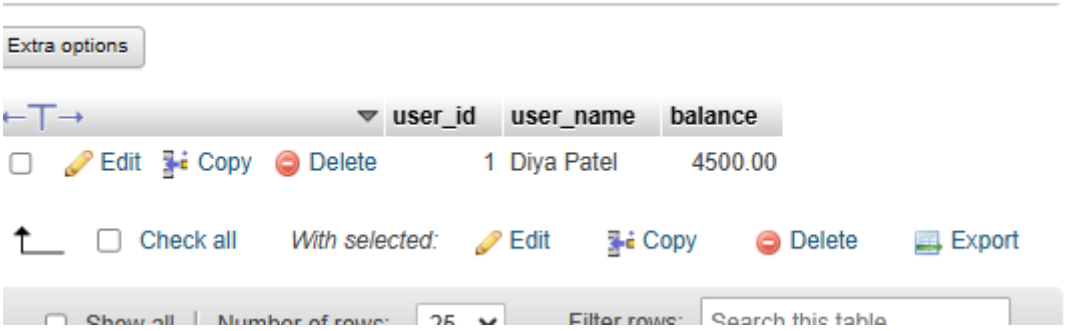
INSERT INTO Transactions

(transaction_id, user_id, amount)

VALUES

(101, 1, 500.00);

SELECT * FROM Wallet;



5.

Use ChatGPT to generate a SQL test scenario where a trigger incorrectly updates a user's balance after an expense entry. Paste the scenario and your corrected trigger code, explaining how you fixed the error.

Hint: Ask ChatGPT for a buggy trigger example and a test case that exposes the bug.

CREATE TABLE WalletTest (

    user_id INT PRIMARY KEY,

```
    balance DECIMAL(10,2)
);

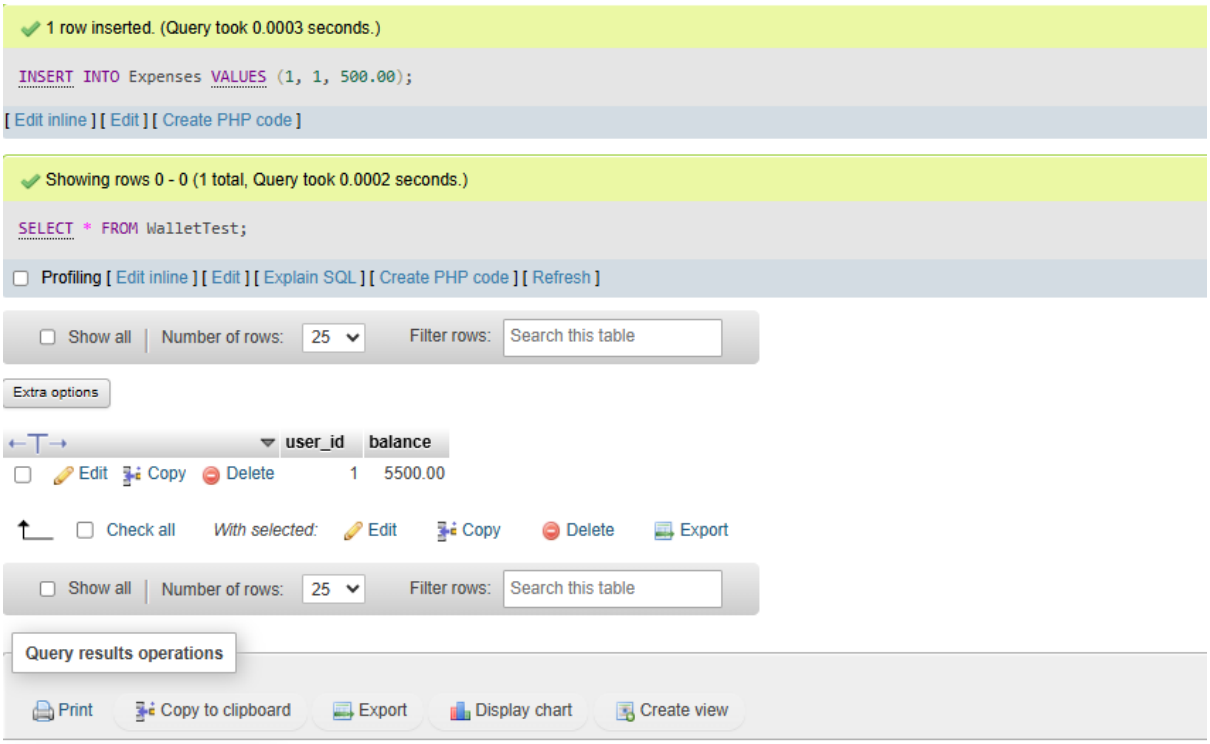
INSERT INTO WalletTest VALUES (1, 5000.00);

CREATE TABLE Expenses (
    expense_id INT PRIMARY KEY,
    user_id INT,
    amount DECIMAL(10,2)
);

CREATE TRIGGER buggy_expense_trigger
AFTER INSERT ON Expenses
FOR EACH ROW
UPDATE WalletTest
SET balance = balance + NEW.amount
WHERE user_id = NEW.user_id;

INSERT INTO Expenses VALUES (1, 1, 500.00);

SELECT * FROM WalletTest;
```



DROP TRIGGER buggy_expense_trigger;

CREATE TRIGGER correct_expense_trigger

AFTER INSERT ON Expenses

FOR EACH ROW

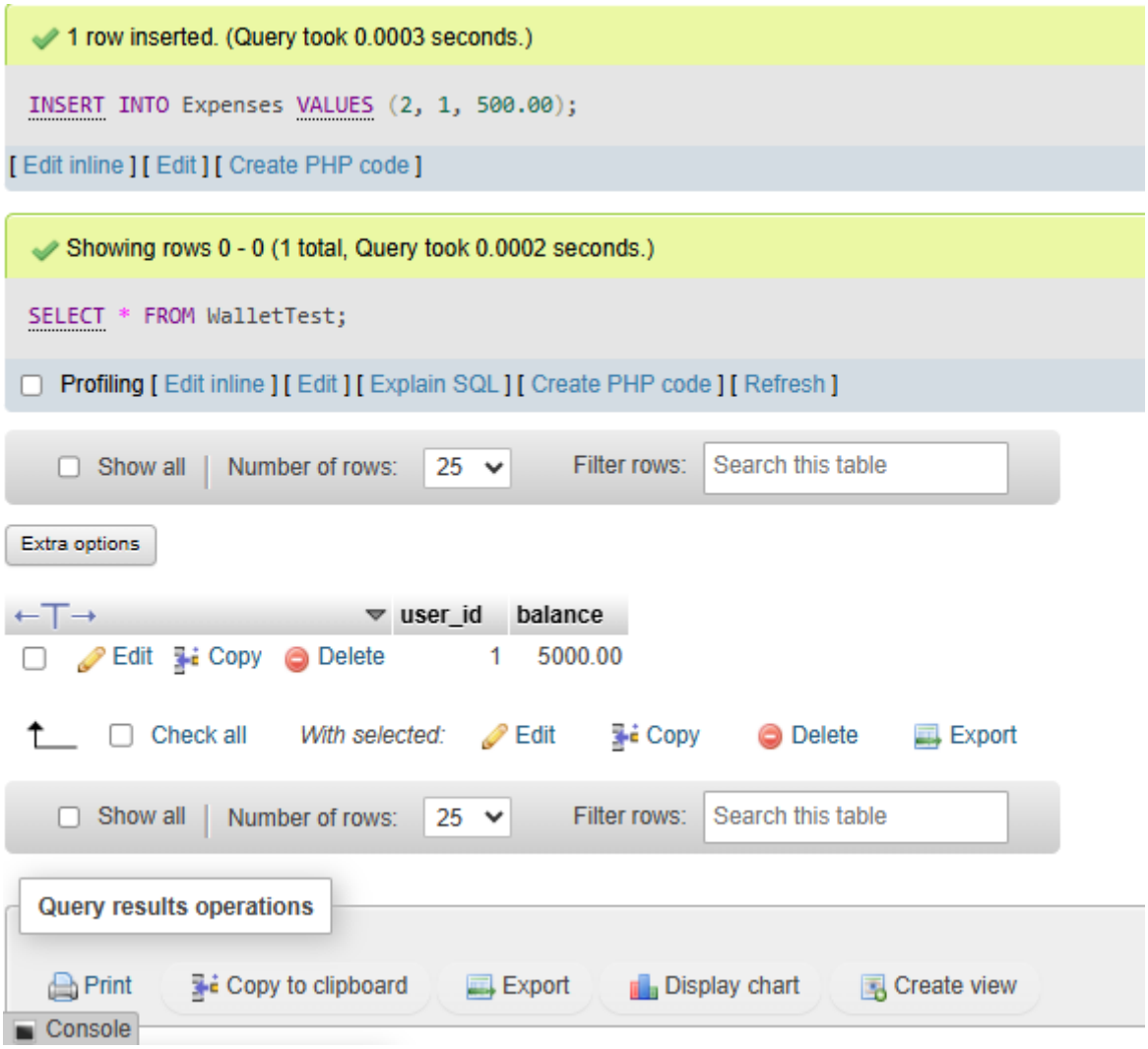
UPDATE WalletTest

SET balance = balance - NEW.amount

WHERE user_id = NEW.user_id;

INSERT INTO Expenses VALUES (2, 1, 500.00);

SELECT * FROM WalletTest;


1 row inserted. (Query took 0.0003 seconds.)
INSERT INTO Expenses VALUES (2, 1, 500.00);
[Edit inline] [Edit] [Create PHP code]
Showing rows 0 - 0 (1 total, Query took 0.0002 seconds.)
SELECT * FROM WalletTest;
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all
Number of rows: 25
Filter rows: Search this table
Extra options
user_id
balance
Edit
Copy
Delete
1
5000.00
Check all
With selected:
Edit
Copy
Delete
Export
Show all
Number of rows: 25
Filter rows: Search this table
Query results operations
Print
Copy to clipboard
Export
Display chart
Create view
Console